

L1-21 LINE RELAY - INSTRUCTION MANUAL (EXCERPT) - PHASE DISTANCE AND POTT FUNCTIONS

1. SAMPLING AND PHASORS

The relay samples VA, VB, VC, IA, IB, IC at 32 samples per power-system cycle (60 Hz, 1920 samples/s). Analog values are in secondary volts and secondary amperes. The oscillography record (COMTRADE 1999, ASCII) stores exactly the samples the protection functions use; the digital channels in the record are the relay's own element outputs at each sample. Sample n in the .dat file (first column, 1-based) is relay sample $k = n - 1$.

For every sample $k \geq 31$ the relay computes, for each analog channel x, the full-cycle Fourier phasor

$$X(k) = (\sqrt{2} / 32) * \text{SUM over } i = 0..31 \text{ of } x[k - i] * \exp(-j * 2 * \pi * (k - i) / 32)$$

(an RMS phasor in a fixed reference frame). No phasor exists before sample 31, and no element can assert before then.

2. PHASE DISTANCE ELEMENTS

Three phase-to-phase loops are measured: AB, BC and CA. For loop XY the loop voltage is $V = V_X - V_Y$ and the loop current is $I = I_X - I_Y$ (phasors at the same sample). A loop is evaluated only when $|I| \geq 50\text{PP}$; its apparent impedance is $Z = V / I$ in secondary ohms. Each zone is a mho circle through the origin (Figure 1). For a forward zone of reach R (setting Z1P or Z2P) and angle T (setting Z1ANG) the loop operates when $|Z - (R/2) * e^{jT}| \leq R/2$. For the reverse zone 3 of reach R (setting Z3P) the loop operates when $|Z + (R/2) * e^{jT}| \leq R/2$. A zone element (Z1, Z2, Z3R) is asserted at sample k when at least one of the three loops operates at sample k.

3. SECONDARY AND PRIMARY QUANTITIES

The relay converts secondary values to primary with its settings CTR and PTR: primary ohms = secondary ohms x PTR / CTR, primary amperes = secondary amperes x CTR. Reaches entered in secondary ohms are applied to the measured secondary impedance directly.

4. FAULT LOCATOR

When the relay trips it reports the distance to fault $d = X / x1$, where X is the reactance of the faulted loop's apparent impedance at the trip sample converted to primary with CTR and PTR, and x1 is the line's positive-sequence reactance per km (0.48 ohm/km for L1).

5. PERMISSIVE OVERREACHING TRANSFER TRIP (POTT) AND ECHO

The scheme logic is shown in Figure 2. Settings EDPU, EDUR and EBLK are in samples. All logic is evaluated once per sample, in the order: zone elements, receive (RX), echo logic, transmit (KEY), trip. The communication channel delivers a KEY asserted at the remote relay at sample k as RX at the local relay at sample $k + 12$ (6.25 ms, each direction). TRIP latches until reset. A breaker opens 96 samples (3 cycles) after its relay asserts TRIP.

Figure 1. Phase mho characteristics (shapes drawn to the Substation B settings).
Forward zones lie on the line angle T ; the reverse zone is the forward circle mirrored through the origin

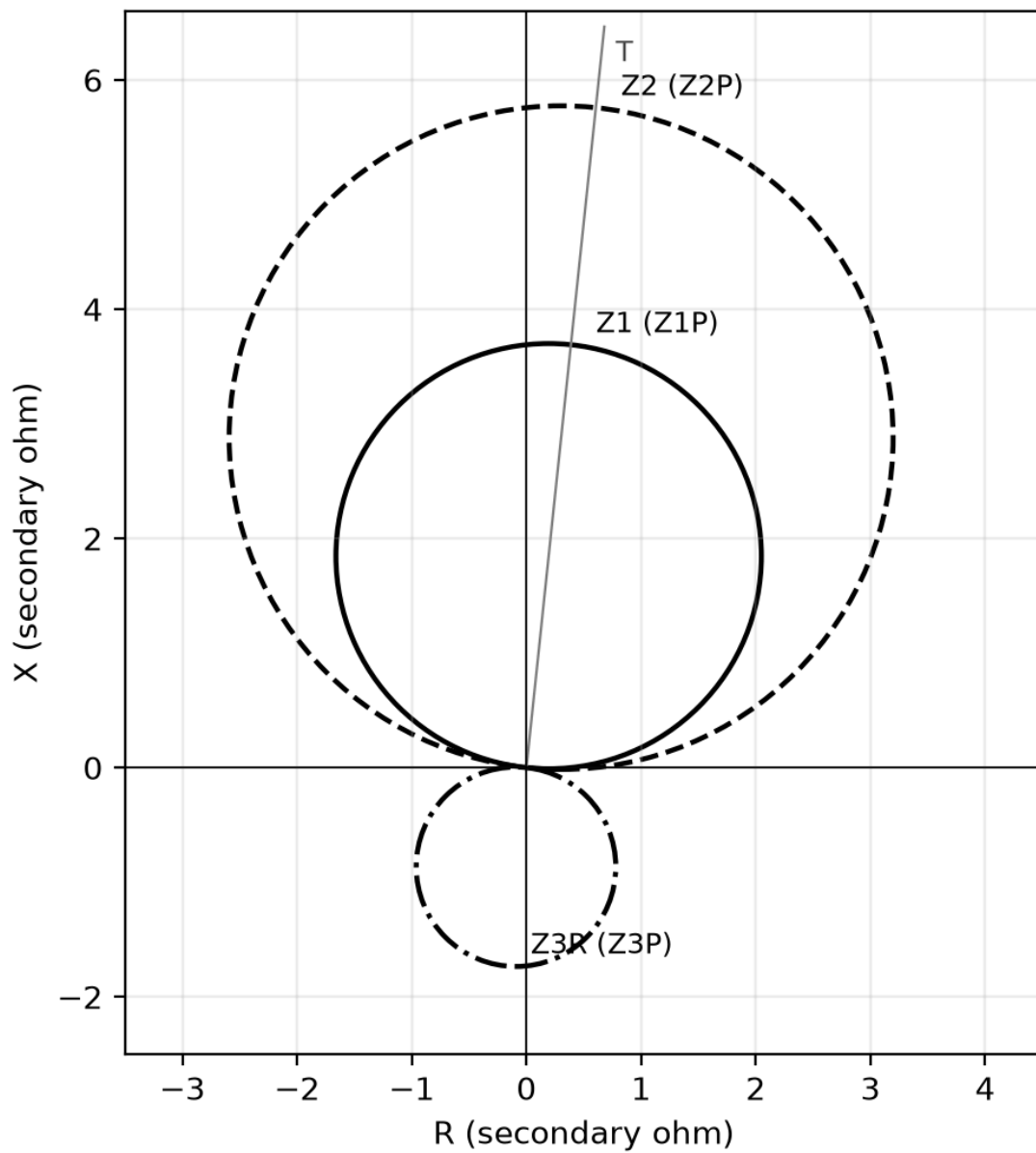
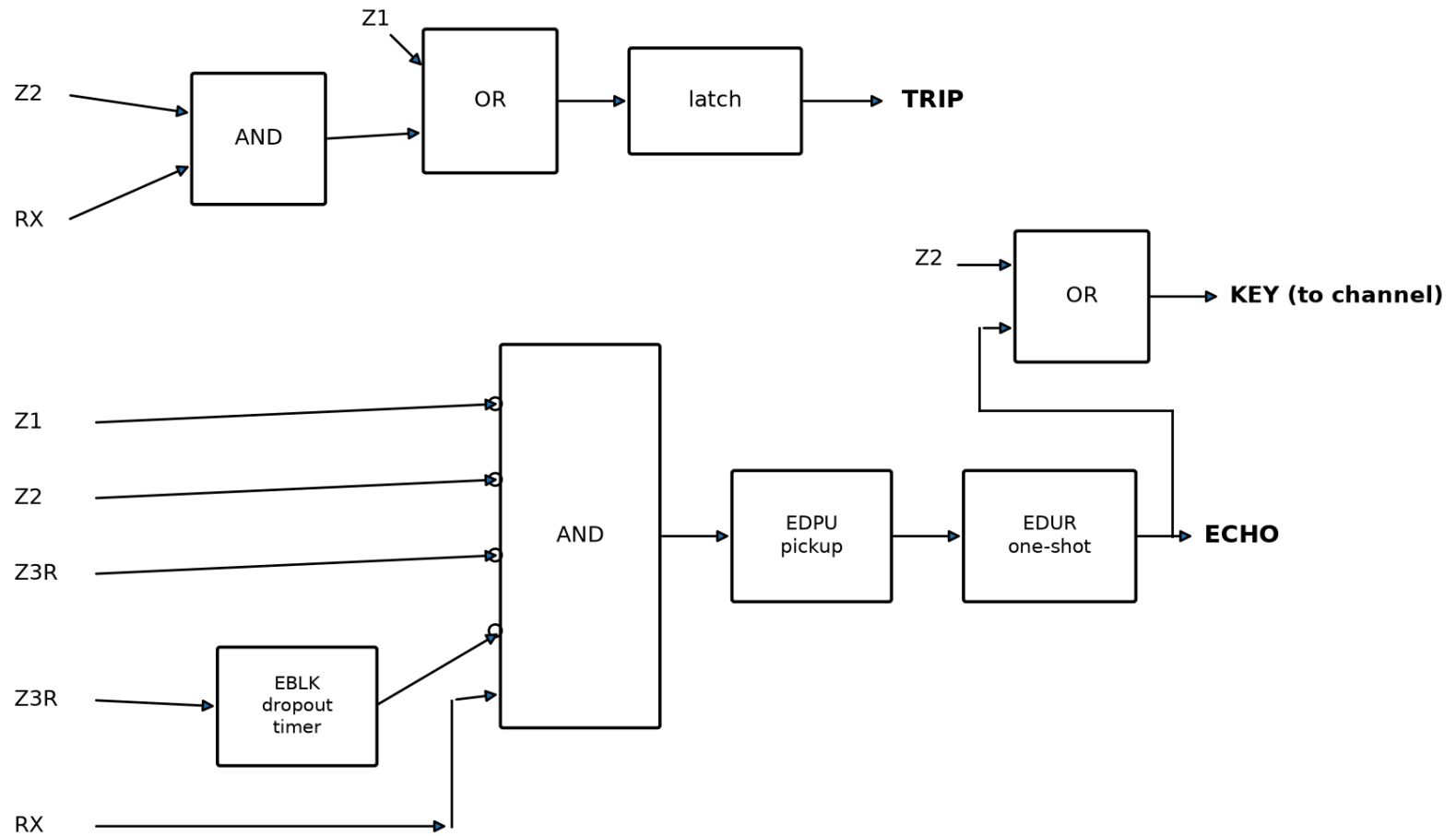


Figure 2. POTT trip, echo and transmit logic (ECHO path active only when setting ECHO = Y)



o = inverted input. EDPU pickup: output asserts on the EDPU-th consecutive sample with the input asserted. EDUR one-shot: output asserted for EDUR samples from a rising input; a new pulse cannot start while one is running. EBLK dropout: output stays asserted for EBLK samples after its input deasserts.